$$A=\int_0^4 (x_R-x_L)\,dx=\int_0^4\left(2\sqrt{y}-\frac{y^2}{4}\right)dy=\left[2\left(\frac{2y^{3/2}}{3}\right)-\frac{y^3}{12}\right]_0^4=\left[2\left(\frac{2\times 8}{3}\right)-\frac{64}{12}\right]-0=\frac{16}{3}\ .$$

Example 9.55

Find the area of the region bounded between the parabola $x^2=y$ and the curve $y=|x|$.

Solution

Both the curves are symmetrical about y-axis.

The curve $y=|x|$ is $y=\begin{cases}x & \text{if } x\geq 0\\ -x & \text{if } x\leq 0\end{cases}$.

It intersects the parabola $x^2=y$ at $(1,1)$ and $(-1,1)$.

The area of the region bounded by the curves is sketched in Fig. 9.27. It lies in the first quadrant as well as in the second quadrant. By symmetry, the required area is twice the area in the first quadrant.

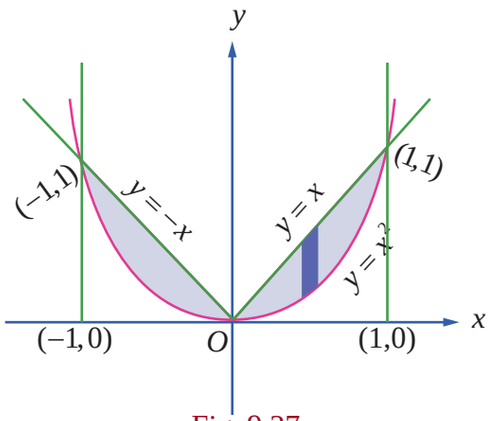


Fig. 9.27

In the first quadrant, the upper curve is $y=x, 0\leq x\leq 1$ and the lower curve is $y=x^2, 0\leq x\leq 1$. Hence, the required area is given by

$$A = 2\int_0^1 [y_U-y_L]\,dx=2\int_0^1[x-x^2]\,dx$$

$$=2\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_0^1$$

$$=2\left(\frac{1}{2}-\frac{1}{3}\right)=\frac{1}{3}.$$

Example 9.56

Find the area of the region bounded by $y=\cos x, y=\sin x$, the lines $x=\frac{\pi}{4}$ and $x=\frac{5\pi}{4}$.

Solution

The region is sketched in Fig. 9.28. The upper boundary of the region is $y=\sin x$ for $\frac{\pi}{4}\leq x\leq\frac{5\pi}{4}$ and the lower boundary of the region is $y=\cos x$ for $\frac{\pi}{4}\leq x\leq\frac{5\pi}{4}$. So the required area A is given by

$$A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}\left(y_U-y_L\right)dx=\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}}(\sin x-\cos x)dx=\left[-\cos x-\sin x\right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$=\left(-\sin\frac{5\pi}{4}-\cos\frac{5\pi}{4}\right)-\left(-\sin\frac{\pi}{4}-\cos\frac{\pi}{4}\right)$$

$$=\left(-\left(-\frac{1}{\sqrt{2}}\right)-\left(-\frac{1}{\sqrt{2}}\right)\right)-\left(-\left(\frac{1}{\sqrt{2}}\right)-\left(\frac{1}{\sqrt{2}}\right)\right)$$

$$=\frac{2}{\sqrt{2}}+\frac{2}{\sqrt{2}}=2\sqrt{2}\ .$$

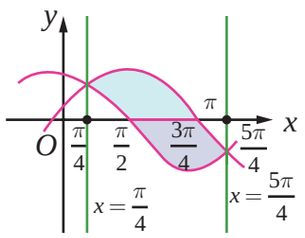


Fig. 9.28

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